

4E1304

Roll No.

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B. Tech. IV - Sem. (Main / Back) Exam., - 2025

Artificial Intelligence and Data Science

4AID3-04 Microprocessor & Interfaces

CS, AID, CAI, CCS, CDS, CIT

Time: 3 Hours

Maximum Marks: 70

Instructions to Candidates:

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL

2. NIL

PART - A

[10×2=20]

(Answer should be given up to 25 words only)

All questions are compulsory

- Q.1 Discuss the concept of static RAM.
- Q.2 Differentiate between microprocessor and microcontroller.
- Q.3 What do you mean by stack pointer?
- Q.4 Define instruction cycle.
- Q.5 Explain the purpose of READY and HOLD pins of 8085 MP.
- Q.6 Draw the flag register of 8085 and explain.
- Q.7 What is WZ pair in 8085 microprocessor?

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- Q.8 Discuss XTHL instruction of 8085 microprocessor.
- Q.9 What are the methods of passing parameters to subroutines?
- Q.10 What is BSR command of 8255 PPI?

PART – B

[5×4=20]

(Analytical/Problem solving questions)

Attempt any five questions

- Q.1/ Discuss the concept of multiplexing and de-multiplexing of buses in 8085 MP.
- Q.2/ What are the different addressing modes in 8085 MP? Discuss each with suitable example.
- Q.3/ Calculate the total delay produced by using an 8-bit register.
- Q.4/ Discuss the concept of RS232C and RS422A.
- Q.5/ Discuss the control word register command for 8254 PIT.
- Q.6/ Discuss the different operating modes used in 8259 PIC.
- Q.7/ Write an assembly language program to add two 8-bit number.

PART – C

[3×10=30]

(Descriptive/Analytical/Problem Solving/Design Questions)

Attempt any three questions

- Q.1 ✓ Draw the architecture of 8085 microprocessor and discuss each part of it.
- Q.2 What are the vectored interrupts? How the address of the interrupt service routine calculated in vectored interrupts? Explain with an example.
- Q.3 ✓ Draw the timing diagram of memory read machine cycle.
- Q.4 ✓ Write the technical note on following -
- (i) IEEE 488 (ii) Liquid crystal display
- Q.5 Write an assembly language program to find the square of a number.